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Fastener chain

Abstract

A fastener chain according to the present invention includes left and right fastener stringers. Each fastener stringer includes a fastener tape and an element row. The element row is formed of fastener elements arranged in a coil shape. The fastener tape includes an element-attached portion and a tape bent portion extending from the element-attached portion and bent in a substantially U-shape. The fastener stringers include respective restraining members in contact with the tape bent portions. Each restraining member is formed by deforming a linear member, and the linear member includes a portion extending in a direction for increasing an overall height dimension of the restraining member. Thus, even when a strong lateral pulling force is applied to the fastener tapes while the element rows are in an engaged state, the element rows can be made less visible from the outside of the fastener.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage application of International Application No. PCT/JP2021/036676, filed Oct. 4, 2021, the contents of which are incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a fastener chain for a concealable fastener.

BACKGROUND ART

(3) An example of a known slide fastener is a slide fastener of a concealable type (hereinafter simply referred to as a concealable slide fastener) in which left and right element rows in an engaged state can be covered with fastener tapes so that the element rows are not visible from the outside. The concealable slide fastener is advantageous in that the element rows are covered to reduce effects on the product's design, and is therefore suitable not only for products such as various clothing and bags but also for seat covers for automobiles and trains.

(4) An example of such a concealable slide fastener is disclosed in International Publication No. 2010/087016 (PTL 1). The concealable slide fastener described in PTL 1 includes restraining members that restrain tape bent portions of left and right fastener tapes from moving away from each other while left and right element rows are engaged and the tape bent portions are in close contact with each other. The restraining members are attached to element-attached portions of the fastener tapes. Each restraining member is positioned such that one side edge thereof is in contact with an inner surface of the corresponding tape bent portion when the element rows are engaged. In addition, each restraining member includes a plurality of rectangular thin-plate-shaped restraining bodies and a film-shaped connecting part that connects the restraining bodies in a tape length direction.

(5) According to the concealable slide fastener of PTL 1, when the left and right fastener tapes receive an outward lateral pulling force along a tape width direction while the left and right element rows are engaged, the left and right tape bent portions are easily maintained in an abutted state. Therefore, satisfactory concealing performance can be obtained.

CITATION LIST

Patent Literature

(6) PTL 1: International Publication No. 2010/087016

SUMMARY OF INVENTION

Technical Problem

(7) The concealability of the concealable slide fastener according to PTL 1 is higher than that of a commonly known concealable slide fastener. However, when, for example, a strong lateral pulling force is applied while the element rows are in the engaged state, the left and right tape bent portions abutted against each other may become separated from each other. As a result, there is a possibility that the element rows in the engaged state concealed behind the tape bent portions will be visible from the outside of the concealable slide fastener.

(8) In addition, according to PTL 1, the restraining bodies of the restraining members attached to the fastener tapes are made of a hard synthetic resin. Therefore, there is also a possibility that the flexibility of the concealable slide fastener will be reduced.

(9) The present invention has been made in light of the above-described problem of the related art, and its object is to provide a fastener chain for a concealable fastener in which element rows are not easily visible from the outside of the fastener even when a strong lateral pulling force is applied to the fastener tapes while the element rows are in an engaged state.

Solution to Problem

(10) To achieve the above-described object, the present invention provides a fastener chain for a

concealable fastener. The fastener chain includes a pair of left and right fastener stringers. Each fastener stringer includes a fastener tape having a first tape surface and a second tape surface and an element row formed on the first tape surface of the fastener tape. The element row is formed of a plurality of fastener elements arranged continuously in a coil shape. The fastener tape includes at least an element-attached portion and a tape bent portion in cross section of the fastener tape. The element-attached portion has the fastener elements attached thereto. The tape bent portion extends from the element-attached portion and is bent in a substantially U-shape such that the second tape surface serves as an inner peripheral surface. The fastener stringers include respective restraining members that restrain the left and right tape bent portions from moving away from each other, the restraining members being in contact with the second tape surfaces of the tape bent portions. Each restraining member is formed by deforming at least one linear member, and includes a portion that the linear member extends in a direction for increasing an overall height dimension of the restraining member.

(11) In the fastener chain according to the present invention, preferably, each restraining member is attached to the element-attached portion of the fastener tape and disposed continuously along a tape length direction of the fastener tape on the second tape surface of the element-attached portion.

(12) In addition, according to the present invention, preferably, each restraining member has a coil shape obtained by helically deforming the linear member.

(13) In this case, preferably, a first core string is inserted between an upper leg portion and a lower leg portion of each fastener element in a length direction of the element row. A second core string is inserted through each restraining member having the coil shape in a length direction of the restraining member. The element row and the restraining member are sewn together on the element-attached portion with a sewing part formed of a plurality of sewing threads. The sewing part extends through the first core string and the second core string.

(14) In addition, according to the present invention, preferably, each restraining member has a shape obtained by deforming the linear member and including a plurality of bent portions bent in an L-shape.

Advantageous Effects of Invention

(15) According to the fastener chain of the present invention, the element rows are not easily visible from the outside of the fastener even when a strong lateral pulling force is applied to the fastener tapes while the element rows are in an engaged state.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic perspective view of part of a fastener chain according to an embodiment of the present invention.

(2) FIG. **2** is a schematic plan view of the fastener chain illustrated in FIG. **1**.

(3) FIG. **3** is a schematic sectional view taken along line III-III in FIG. **2**.

(4) FIG. **4** is a schematic side view of the fastener chain illustrated in FIG. **1**.

(5) FIG. **5** illustrates a fastener tape, a fastener element, and a restraining member subjected to a sewing process.

(6) FIG. **6** is a schematic perspective view of part of a fastener chain according to a modification.

(7) FIG. **7** is a schematic plan view of the fastener chain illustrated in FIG. **6**.

(8) FIG. **8** is a schematic sectional view taken along line VIII-VIII in FIG. **7**.

(9) FIG. **9** is a schematic side view of the fastener chain illustrated in FIG. **6**.

DESCRIPTION OF EMBODIMENTS

(10) A preferred embodiment of the present invention will now be described in detail with reference to the drawings. The present invention is not limited to the embodiment described below

in any way, and various changes are possible as long as a configuration substantially the same as that of the present invention is provided and operational effects similar to those of the present invention are obtained.
(11) FIG. **1** is a schematic perspective view of a fastener chain for a concealable fastener according to the present embodiment. FIGS. **2** and **4** are respectively a schematic plan view and a schematic side view of part of the fastener chain. FIG. **3** is a schematic sectional view taken along line III-III in FIG. **2**. FIGS. **1** and **2** and FIGS. **6** and **7** illustrating a modification described below each show a fastener chain from which fastener tapes are partially removed to facilitate understanding of the structure of the fastener chain.
(12) In the following description, a front-rear direction is a length direction of a fastener chain **10** or fastener tapes **12**, for example, a direction in which an opening-closing member (or slider) is moved to cause left and right element rows **20** to engage with or separate from each other (see FIGS. **2** and **4**). Specifically, the front-rear direction is the up-down direction in the plan view of FIG. **2**.
(13) The up-down direction is a direction orthogonal to the front-rear direction and along a thickness direction of the fastener chain **10** (see FIGS. **3** and **4**). In this case, the direction from fastener elements **21** toward the fastener tapes **12** and restraining members **30** is defined as upward, and the direction opposite thereto is defined as downward. A left-right direction is a direction orthogonal to the front-rear direction and the up-down direction, and is a direction along which the fastener elements **21** move when the element rows **20** are engaged or separated (see FIGS. **2** and **3**). Specifically, the up-down direction and the left-right direction are respectively the up-down direction and the left-right direction in the sectional view of FIG. **3**.
(14) The concealable fastener according to the present embodiment includes the fastener chain **10** including a pair of left and right fastener stringers **11** and a dedicated opening-closing member (referred to also as a jig) that is not illustrated. The opening-closing member is attachable to and removable from the left and right element rows **20** of the fastener chain **10**. In this concealable fastener, the left and right element rows **20** can be engaged with and separated from each other by attaching the opening-closing member (not illustrated) to the element rows **20** and moving the opening-closing member along the element rows **20**.
(15) In the present invention, the structure of the opening-closing member is not particularly limited as long as the opening-closing member is attachable to and removable from the left and right element rows **20** and capable of causing the left and right element rows **20** to engage with and separate from each other. In addition, in the present invention, instead of using the opening-closing member, a slider capable of causing the left and right element rows of the fastener chain **10** to engage with and separate from each other may be attached to the element rows **20** in a slidable manner to form the concealable slide fastener.
(16) The fastener chain **10** of the present embodiment includes the pair of left and right fastener stringers **11**. Each of the left and right fastener stringers **11** includes the fastener tape **12** having a first tape surface **12***a* and a second tape surface **12***b*; the element row **20** provided on the first tape surface **12***a* of the fastener tape **12**; and the restraining member **30** provided on the second tape surface **12***b* of the fastener tape **12** and attached to the fastener tape **12**. The fastener tape **12** has the shape of a thin, elongated strip. The fastener tape **12** has the first tape surface **12***a* and the second tape surface **12***b* on opposite sides in a tape thickness direction.
(17) In a cross section orthogonal to a tape length direction of the fastener tapes **12** (see FIG. **3**), each fastener tape **12** includes an element-attached portion **13** to which the element row **20** is attached; a tape bent portion **14** that extends from the element-attached portion **13** and that is bent in a substantially U-shape; and a tape extending portion **15** that extends further from the tape bent portion **14**. The fastener tape **12** has this cross-sectional shape over the entirety thereof in the length direction of the fastener tape **12**.
(18) The first tape surface **12***a* of each fastener tape **12** forms an outer peripheral surface of the

substantially U-shaped tape bent portion **14**, and the second tape surface **12***b* of each fastener tape **12** forms an inner peripheral surface of the tape bent portion **14**. In the fastener chain **10** in which the left and right element rows **20** are engaged, the first tape surfaces **12***a* of portions of the tape bent portions **14** and the tape extending portions **15** serve as chain outer surfaces exposed to the outside of the fastener chain **10**.

(19) The element-attached portion **13** of each fastener tape **12** is provided on one side edge of the fastener tape **12**. The element-attached portions **13** have the fastener elements **21** constituting the element rows **20** and the restraining members **30** sewn thereon. The tape bent portion **14** of each fastener tape **12** is abutted against the tape bent portion **14** of the other fastener tape **12** when the left and right element rows **20** are engaged, so that the element rows **20** in the engaged state are covered by the fastener tapes **12** and not visible from the outside. The tape extending portions **15** extend outward from the tape bent portions **14** in the left-right direction so as to cover the restraining members **30**.

(20) According to the fastener chain **10** of the present embodiment, a fastener-receiving member, such as a seat cover for an automobile, is fixed to a portion of the tape bent portion **14** and the tape extending portion **15** of each fastener tape **12**, preferably a portion of the tape bent portion **14**, by sewing. When the fastener-receiving member is fixed to the tape bent portion **14** of each fastener tape **12** as described above, the fastener chain **10** can be made non-visible or less visible from the outside of the fastener-receiving member when the left and right element rows **20** are engaged.

(21) Each of the left and right element rows **20** is formed of the fastener elements **21** arranged continuously. The continuous fastener elements **21** are formed by deforming a single monofilament made of a synthetic resin, such as polyamide or polyester, into a coil shape and pressing the monofilament at predetermined intervals to form engaging heads **22** including bulging portions that partially bulge in the front-rear direction.

(22) The fastener elements **21** are formed substantially similarly to commonly known fastener elements formed in a coil shape. More specifically, each fastener element **21** includes the engaging head **22** that engages with the fastener element **21** to be engaged therewith; an upper leg portion **23** and a lower leg portion **24** respectively extending from an upper end and a lower end of the engaging head **22**; and a connecting portion **25** that connects an end of the upper leg portion **23** or the lower leg portion **24** to the lower leg portion **24** or the upper leg portion **23** of the fastener element **21** adjacent thereto in the front-rear direction. In this case, the upper leg portion **23** of the fastener element **21** is in contact with the fastener tape **12**, and the lower leg portion **24** is separated from the fastener tape **12**.

(23) As illustrated in FIG. **3**, the fastener elements **21** are fixed to the first tape surface **12***a* of the element-attached portion **13** of each fastener tape **12** by a sewing part **50** together with a first core string **41** while the first core string **41** is inserted between the upper leg portion **23** and the lower leg portion **24** of each fastener element **21** in the front-rear direction or the length direction of each element row **20**. Each fastener element **21** is fixed to the fastener tape **12** such that the engaging head **22** thereof projects outward from the position of the tape bent portion **14** of the fastener tape **12** in the left-right direction. Each sewing part **50** is formed in a sewing process described below by performing double-chain-stitch sewing using a plurality of sewing threads. The sewing parts **50** are shown only in FIG. **3**, and are omitted in FIGS. **1**, **2**, and **4** to facilitate understanding of the relationship between the element rows **20** and the restraining members **30**.

(24) Each restraining member **30** is coil-shaped. The restraining member **30** is formed by deforming a monofilament made of a synthetic resin, which is a single elongated linear member, into a regular helical shape. Here, the linear member is an elongated member having a constant or substantially constant shape in cross section orthogonal to the length direction. The restraining member **30** has a substantially elliptical outline in a cross section orthogonal to the front-rear direction (FIG. **3**). As illustrated in FIG. **4**, for example, the restraining member **30** of the present embodiment is formed such that the pitch of the coil shape in the front-rear direction is equal or

substantially equal to the pitch of the fastener elements **21** in the front-rear direction.
(25) In the above-described cross section, each coil-shaped restraining member **30** includes: an upper curved portion **31** that convexly curves upward; a left curved portion **32** that extends from a left end of the upper curved portion **31** and convexly curves leftward; a lower curved portion **33** that extends from a lower end of the left curved portion **32** and convexly curves downward; and a right curved portion **34** that extends from a right end of the lower curved portion **33** to be connected to the next upper curved portion **31** and convexly curves rightward.
(26) In this case, the left curved portion **32** and the right curved portion **34** of the restraining member **30** serve as portions of the restraining member **30** in which the monofilament extends in a direction for increasing the overall height dimension of the restraining member **30**. Here, the height dimension is the dimension in the up-down direction. In the present embodiment, the upper curved portion **31**, the left curved portion **32**, the lower curved portion **33**, and the right curved portion **34** of the restraining member **30** are, for example, sectioned from each other at boundaries provided at positions at which the tangent line of the outer peripheral surface of the restraining member **30** having a substantially elliptical shape is at an angle of 45° with respect to the up-down or left-right direction in the cross section of the restraining member **30**.
(27) While each coil-shaped restraining member **30** is in a state such that a second core string **42** is inserted therethrough in the front-rear direction, the restraining member **30** is fixed to the second tape surface **12***b* of the element-attached portion **13** of the fastener tape **12** with the sewing part **50** together with the second core string **42**. The restraining member **30** extends continuously in the front-rear direction over the entirety of the fastener tape **12** in the front-rear direction.
(28) In the present embodiment, a twine or a knit cord obtained by knitting around a bundle of threads, for example, may be suitably used as the first core string **41** and the second core string **42**. In the present invention, the material of the first core string **41** and the second core string **42** is not particularly limited. In FIGS. **1** to **4**, the second core string **42** is inserted through each coil-shaped restraining member **30** with no gap therebetween. However, in the present invention, a gap may be provided between the restraining member **30** and the second core string **42** as long as the second core string **42** can be sewn on and fixed to the fastener tape **12** together with the restraining member **30**.
(29) In the present embodiment, a maximum dimension Wm of the entirety of each restraining member **30** in the left-right direction is greater than a maximum dimension of the first core string **41** in the left-right direction and less than a maximum dimension of the entirety of each fastener element **21** in the left-right direction. In addition, the restraining member **30** is disposed in a region between an outer edge of the engaging head **22** of each fastener element **21** and an outer edge of the connecting portion **25** of the fastener element **21** in the left-right direction.
(30) A maximum dimension Hm of the entirety of each restraining member **30** in the up-down direction is greater than the thickness of the fastener tape **12** and the thickness of the monofilament forming the fastener elements **21**.
(31) Each coil-shaped restraining member **30** is flexible, and is easily bendable upward, downward, leftward, and rightward with respect to the front-rear direction. The restraining member **30** sewn on the element-attached portion **13** of the fastener tape **12** is in contact with the second tape surface **12***b* of the tape bent portion **14** of the fastener tape **12**, so that the tape bent portion **14** is restrained from moving away from the tape bent portion **14** of the fastener stringer **11** engaged therewith. The restraining member **30** may be of the same size as the fastener elements **21**, or larger than the fastener elements **21**. The restraining member **30** may be identical to the fastener elements **21**.
(32) To manufacture the fastener chain **10** of the present embodiment described above, first, the following are prepared: the elongated strip-shaped fastener tape **12**; the fastener elements **21** that are continuous and through which the first core string **41** is inserted; and the coil-shaped restraining member **30** through which the second core string **42** is inserted.
(33) Next, as illustrated in FIG. **5**, the fastener tape **12** is held flat. In addition, the fastener elements

21 through which the first core string **41** is inserted are brought into contact with the first tape surface **12***a* of the flat fastener tape **12**, and the restraining member **30** through which the second core string **42** is inserted is brought into contact with the second tape surface **12***b* of the fastener tape **12**. At this time, the fastener elements **21** are brought into contact with the element-attached portion **13** of the fastener tape **12** while the engaging heads **22** thereof face toward the tape extending portion **15** of the fastener tape **12**. In addition, the restraining member **30** is brought into contact with the element-attached portion **13** of the fastener tape **12** such that the second core string **42** and the first core string **41** inserted through the fastener elements **21** overlap in the up-down direction.

(34) Subsequently, the fastener elements **21**, the restraining member **30**, and the fastener tape **12** are set to a sewing machine and subjected to a sewing process while the fastener elements **21** and the restraining member **30** are in contact with the fastener tape **12** as described above. In the sewing process, double-chain-stitch sewing is performed by moving a sewing needle upward and downward so that the sewing needle penetrates through the second core string **42**, the fastener tape **12**, and the first core string **41**. Thus, the sewing part **50** can be formed by using sewing threads in a single sewing process performed by the sewing machine, and the fastener elements **21** and the restraining member **30** can be sewn together and fixed to the element-attached portion **13** of the fastener tape **12** with the sewing part **50**. In this case, the sewing part **50** formed by the sewing machine extends through the second core string **42**, the fastener tape **12**, and the first core string **41**. In the present invention, the sewing part **50**, with which the fastener elements **21**, the fastener tape **12**, and the restraining member **30** are fixed, may be formed of a stitch other than the double-chain stitch, such as a lock stitch.

(35) In the present embodiment, after the fastener elements **21** and the restraining member **30** are fixed to the fastener tape **12** by a single sewing process as described above, the fastener tape **12** that has been held flat is bent into a U-shape with the second tape surface **12***b* serving as an inner peripheral surface so that the fastener tape **12** covers the restraining member **30**. Thus, the U-shaped tape bent portion **14** of the fastener tape **12** is formed, and the engaging heads **22** of the fastener elements **21** project beyond the position of the first tape surface **12***a* of the tape bent portion **14** in the left-right direction.

(36) After that, the fastener tape **12** is subjected to heat setting while the fastener tape **12** is in the bent state, so that the shape of the fastener tape **12** is stabilized.

(37) Thus, the fastener stringer **11** according to the present embodiment is manufactured. Two fastener stringers **11** are manufactured as a pair, and the element rows **20** thereof are engaged with each other. Thus, the fastener chain **10** according to the present embodiment illustrated in FIGS. **1** to **4** is manufactured.

(38) When the manufactured fastener chain **10** is attached to, for example, a seat cover (not illustrated) of an automobile that serves as a fastener-receiving member, at least a portion of the tape bent portion **14** and the tape extending portion **15** of each fastener tape **12**, preferably at least a portion of the tape bent portion **14**, is caused to overlap the seat cover, and the overlapping portions are subjected to a sewing process so that the fastener tape **12** is sewn on the seat cover. Thus, the seat cover to which the fastener chain **10** is attached can be obtained. The fastener chain **10** of the present embodiment may also be used for products other than the seat cover, such as clothing and bags.

(39) In the above-described fastener chain **10** of the present embodiment, when the left and right element rows **20** are engaged with each other, the tape bent portions **14** of the left and right fastener tapes **12** are abutted against each other. In addition, the coil-shaped restraining members **30** fixed to the element-attached portions **13** of the fastener tapes **12** are in contact with the second tape surfaces **12***b* of the tape bent portions **14** of the fastener tapes **12**, so that the left and right tape bent portions **14** are maintained abutted against each other in close contact with each other. Accordingly, the left and right element rows **20** in the engaged state are covered by the left and right fastener

tapes **12** and are not visible from the outside of the fastener chain **10** (from above in FIG. **3**).

(40) In the fastener chain **10** in which the left and right element rows **20** are engaged, the fastener tapes **12** may, for example, receive a tensile force or a lateral pulling force from the seat cover, which is the fastener-receiving member, in a direction for separating the left and right tape bent portions **14** from each other. In such a case, since the left and right coil-shaped restraining members **30** are in contact with the respective tape bent portions **14** to restrain the tape bent portions **14** from moving away from each other, the left and right tape bent portions **14** can be easily maintained in close contact with each other as described above.

(41) Furthermore, if the left and right fastener tapes **12** receive a very strong tensile force or lateral pulling force, for example, there is a possibility that the left and right tape bent portions **14** will be separated from each other even when the tape bent portions **14** are restrained from moving by the coil-shaped restraining members **30**. However, in the fastener chain **10** of the present embodiment, since the restraining members **30** are coil-shaped, the overall height dimension of the restraining members **30** is greater than, for example, that of the restraining members described in PTL 1. Therefore, the left and right element rows **20** in the engaged state are spaced by a large distance from the tape surfaces of the fastener tapes **12** that constitute the outer surface of the fastener chain **10**.

(42) As a result, even when the left and right fastener tapes **12** receive a strong tensile force or lateral pulling force as described above and the left and right tape bent portions **14** move against the restraining force applied by the restraining members **30** such that a gap is formed therebetween (that is, even when the left and right fastener tapes **12** are released from the close contact state), the left and right element rows **20** are spaced from the outer surface of the fastener chain **10** in the up-down direction. Therefore, the left and right element rows **20** are not easily visible from the outside of the fastener chain **10** through the gap between the left and right tape bent portions **14**.

(43) In particular, in this case, since the element rows **20** are spaced by a large distance from the outer surface of the fastener chain **10**, even when a gap is formed between the left and right tape bent portions **14**, the gap between the tape bent portions **14** appears dark because light does not easily reach the element rows **20**. As a result, the left and right element rows **20** can be made less visible.

(44) Thus, according to the fastener chain **10** for the concealable fastener of the present embodiment, when the fastener tapes **12** receive a tensile force or a lateral pulling force, the left and right element rows **20** can be made less visible compared to those in the concealable slide fastener according to the related art described in PTL 1. Therefore, the concealability of the element rows **20** can be increased. Accordingly, degradation of the appearance quality or design of the product to which the fastener chain **10** is attached due to exposure of the element rows **20** can be suppressed.

(45) In addition, in the fastener chain **10** of the present embodiment, since the restraining members **30** are coil-shaped, each of the left and right fastener stringers **11** can be highly flexible. Therefore, the reduction in flexibility of the fastener chain **10** due to the installation of the restraining members **30** can be suppressed, so that the fastener chain **10** can be easily bent in the up-down direction and the left-right direction, in particular, in the left-right direction, with respect to the front-rear direction.

(46) Thus, the fastener chain **10** of the present embodiment can be easily attached to the fastener-receiving member, such as the seat cover, while the fastener chain **10** is curved in, for example, the left-right direction. In addition, when the fastener chain **10** is attached to the fastener-receiving member in, for example, an arc shape curved in the left-right direction, the fastener chain **10** can be attached to the fastener-receiving member in an arc shape with a smaller radius of curvature compared to the known slide fastener. As a result, the range of uses of the fastener chain **10** can be expanded, and the appearance quality and design of the product to which the fastener chain **10** is attached can be improved.

(47) In the fastener chain **10** of the above-described embodiment, the restraining members **30** are coil-shaped, and the second core string **42** is inserted through each restraining member **30**. However, in the present invention, the form of the restraining member for restricting the movement of the tape bent portion **14** of each fastener tape **12** is not limited to this. The restraining member of the present invention may be any restraining member that is formed by deforming a linear member and in which the linear member includes a portion that extends in a direction for increasing the overall height dimension of the restraining member. For example, the restraining member according to the present invention may have a zigzag shape including a plurality of bent portions bent in an L-shape as in a fastener chain **10***a* according to a modification illustrated in FIGS. **6** to **9**.
(48) More specifically, zigzag-shaped restraining members **60** illustrated in FIGS. **6** to **9** are each shaped such that first columnar portions **61** extending in the up-down direction, second columnar portions **62** extending in the left-right direction, and third columnar portions **63** extending in the front-rear direction are regularly arranged. Connecting portions connecting the different types of columnar portions, that is, the first columnar portions **61**, the second columnar portions **62**, and the third columnar portions **63**, are formed of bent portions **64** that are bent in an L-shape.
(49) Each zigzag-shaped or L-shaped restraining member **60** is formed by deforming a monofilament made of a synthetic resin, which is a single elongated linear member, to form the above-described L-shaped bent portions **64** on the monofilament. In this case, the first columnar portions **61** extending in the up-down direction serve as portions of the restraining member **60** in which the monofilament extends in the direction for increasing the overall height dimension of the restraining member **60**.
(50) As illustrated in FIG. **8**, each zigzag-shaped restraining member **60** does not have the second core string **42** of the above-described embodiment inserted between the upper and lower second columnar portions **62** extending in the left-right direction, and only the zigzag-shaped restraining member **60** is sewn on and fixed to the second tape surface **12***b* of the element-attached portion **13** of the fastener tape **12**. In addition, each zigzag-shaped restraining member **60** is fixed to the element-attached portion **13** together with the fastener elements **21** by a sewing part **70**. Each sewing part **70** is formed in a sewing process by performing double-chain-stitch sewing using a plurality of sewing threads. In this modification, the sewing parts **70** are shown only in FIG. **8**, and are omitted in FIGS. **6**, **7**, and **9**.
(51) The fastener chain **10***a* according to this modification is formed substantially similarly to the fastener chain **10** according to the embodiment except that the zigzag-shaped restraining members **60** are used instead of the restraining members **30** and the second core strings **42** in the fastener chain **10** according to the above-described embodiment. Therefore, members and components of the fastener chain **10***a* of this modification that are substantially similar to those in the embodiment are denoted by the same reference signs, and description thereof will be omitted. In this modification, each zigzag-shaped restraining member **60** may be sewn on the element-attached portion **13** of the fastener tape **12** by the sewing part **70** together with a second core string while the second core string is inserted between the upper and lower second columnar portions **62** in the front-rear direction.
(52) Each zigzag-shaped restraining member **60** according to the modification is flexible, and is easily bendable upward, downward, leftward, and rightward with respect to the front-rear direction. The zigzag-shaped restraining member **60** is fixed to the element-attached portion **13** while being in contact with the second tape surface **12***b* of the tape bent portion **14** of the fastener tape **12**. Therefore, the zigzag-shaped restraining member **60** restrains the tape bent portion **14** from moving away from the tape bent portion **14** of the fastener stringer **11** engaged therewith.
(53) The fastener chain **10***a* including the above-described zigzag-shaped restraining members **60** also provides effects similar to those of the fastener chain **10** including the coil-shaped restraining members **30** of the above-described embodiment.
(54) In the above-described embodiment and modification, the restraining members **30** and **60** are

formed by deforming a monofilament made of a synthetic resin. However, in the present invention, the material of the restraining members **30** and **60** is not particularly limited. Each restraining member of the present invention may be formed by, for example, deforming a metal linear member, such as a wire, into a coil shape or a zigzag shape including a plurality of bent portions bent in an L-shape. Alternatively, each restraining member may be formed of, for example, a metal coil spring.

REFERENCE SIGNS LIST

(55) **10**, **10***a* fastener chain **11** fastener stringer **12** fastener tape **12***a* first tape surface **12***b* second tape surface **13** element-attached portion **14** tape bent portion **15** tape extending portion **20** element row **21** fastener element **22** engaging head **23** upper leg portion **24** lower leg portion **25** connecting portion **30** restraining member **31** upper curved portion **32** left curved portion **33** lower curved portion **34** right curved portion **41** first core string **42** second core string **50** sewing part **60** restraining member **61** first columnar portion **62** second columnar portion **63** third columnar portion **64** bent portion **70** sewing part Hm maximum dimension of entirety of restraining member in up-down direction Wm maximum dimension of entirety of restraining member in left-right direction

Claims

1. A fastener chain for a concealable fastener, the fastener chain including a pair of left and right fastener stringers, each fastener stringer including a fastener tape having a first tape surface and a second tape surface and an element row formed on the first tape surface of the fastener tape, the element row being formed of a plurality of fastener elements arranged continuously in a coil shape, the fastener tape including at least an element-attached portion and a tape bent portion in cross section of the fastener tape, the element-attached portion having the fastener elements attached thereto, the tape bent portion extending from the element-attached portion and being bent in a substantially U-shape such that the second tape surface serves as an inner peripheral surface, wherein the fastener stringers include respective restraining members that restrain the left and right tape bent portions from moving away from each other, the restraining members being in contact with the second tape surfaces of the tape bent portions; and each restraining member is formed by deforming at least one linear member, and includes a portion that the linear member extends in a direction for increasing an overall height dimension of the restraining member.
2. The fastener chain according to claim 1, wherein each restraining member is attached to the element-attached portion of the fastener tape and disposed continuously along a tape length direction of the fastener tape on the second tape surface of the element-attached portion.
3. The fastener chain according to claim 1, wherein each restraining member has a coil shape obtained by helically deforming the linear member.
4. The fastener chain according to claim 3, wherein a first core string is inserted between an upper leg portion and a lower leg portion of each fastener element in a length direction of the element row, wherein a second core string is inserted through each restraining member having the coil shape in a length direction of the restraining member, wherein the element row and the restraining member are sewn together on the element-attached portion with a sewing part formed of a plurality of sewing threads, and wherein the sewing part extends through the first core string and the second core string.
5. The fastener chain according to claim 1, wherein each restraining member has a shape obtained by deforming the linear member and including a plurality of bent portions bent in an L-shape.